

as a standalone unit. You can run the board from a 9V alkaline battery or a 7V-12 DC power supply using a standard 2.1mm male DC Jack plug.

This circuit can also be used to drive 230V AC mains input lamps instead of LEDs. For this, you need to assemble four solidstate light switch modules

with galvanic isolation (and zero-crossing) as shown in Fig. 5.

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FANCY_LIGHTS.PDE

```
// Fancy Lights Controller //
int switchPin = 2; // mode switch is
connected to pin 2
int led1Pin = 12;
int led2Pin = 11;
int led3Pin = 10;
int led4Pin = 9;
int val; // variable for reading the
pin status
int val2; // variable for reading the
delayed status
int buttonState; // variable to hold the
mode switch state
int lightMode = 0;
void setup() {
  pinMode(switchPin, INPUT); // Set the
mode switch pin as input
  pinMode(led1Pin, OUTPUT);
  pinMode(led2Pin, OUTPUT);
  pinMode(led3Pin, OUTPUT);
  pinMode(led4Pin, OUTPUT);
  Serial.begin(9600); // Set up serial
communication at 9600bps
  buttonState = digitalRead(switchPin); //
read the initial state
}
void loop() {
  val = digitalRead(switchPin); // read
input value and store it in val
  delay(10);
  val2 = digitalRead(switchPin); // Key
Bounce Prevention
  if (val == val2) {
    if (val != buttonState) {
      if (val == HIGH) { // check if the mode
        button is pressed
        if (lightMode == 0) {
          lightMode = 1; // turn LEDs on
        }
        else {
          if (lightMode == 1) { // if LEDs on
            lightMode = 2; // make it flash
          }
          else {
            if (lightMode == 2) { // if LEDs flashing
              lightMode = 3; // make it wave
            }
            else {
              if (lightMode == 3) { // if LEDs waving,
                lightMode = 0; // turn LEDs off
              }
            }
          }
        }
        buttonState = val; // save the new state
in variable
        // Now do whatever the lightMode indi-
cates
        if (lightMode == 0) { // LEDs-off
          digitalWrite(led1Pin, LOW);
          digitalWrite(led2Pin, LOW);
          digitalWrite(led3Pin, LOW);
          digitalWrite(led4Pin, LOW);
        }
        // T.K.Hareendran
        (7)
        if (lightMode == 1) { // LEDs-on
          digitalWrite(led1Pin, HIGH);
          digitalWrite(led2Pin, HIGH);
          digitalWrite(led3Pin, HIGH);
          digitalWrite(led4Pin, HIGH);
        }
        if (lightMode == 2) { // LEDs-flashing
          digitalWrite(led1Pin, HIGH);
          digitalWrite(led2Pin, HIGH);
          digitalWrite(led3Pin, HIGH);
          digitalWrite(led4Pin, HIGH);
          delay(100);
          digitalWrite(led1Pin, LOW);
          digitalWrite(led2Pin, LOW);
          digitalWrite(led3Pin, LOW);
          digitalWrite(led4Pin, LOW);
          delay(100);
        }
        if (lightMode == 3) { // LEDs-waving
          digitalWrite(led4Pin, LOW);
          digitalWrite(led1Pin, HIGH);
          digitalWrite(led2Pin, HIGH);
          delay(50);
          digitalWrite(led2Pin, LOW);
          digitalWrite(led3Pin, HIGH);
          delay(50);
          digitalWrite(led3Pin, LOW);
          digitalWrite(led4Pin, HIGH);
          delay(50);
          digitalWrite(led4Pin, LOW);
        }
      }
    }
  }
}
```

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